

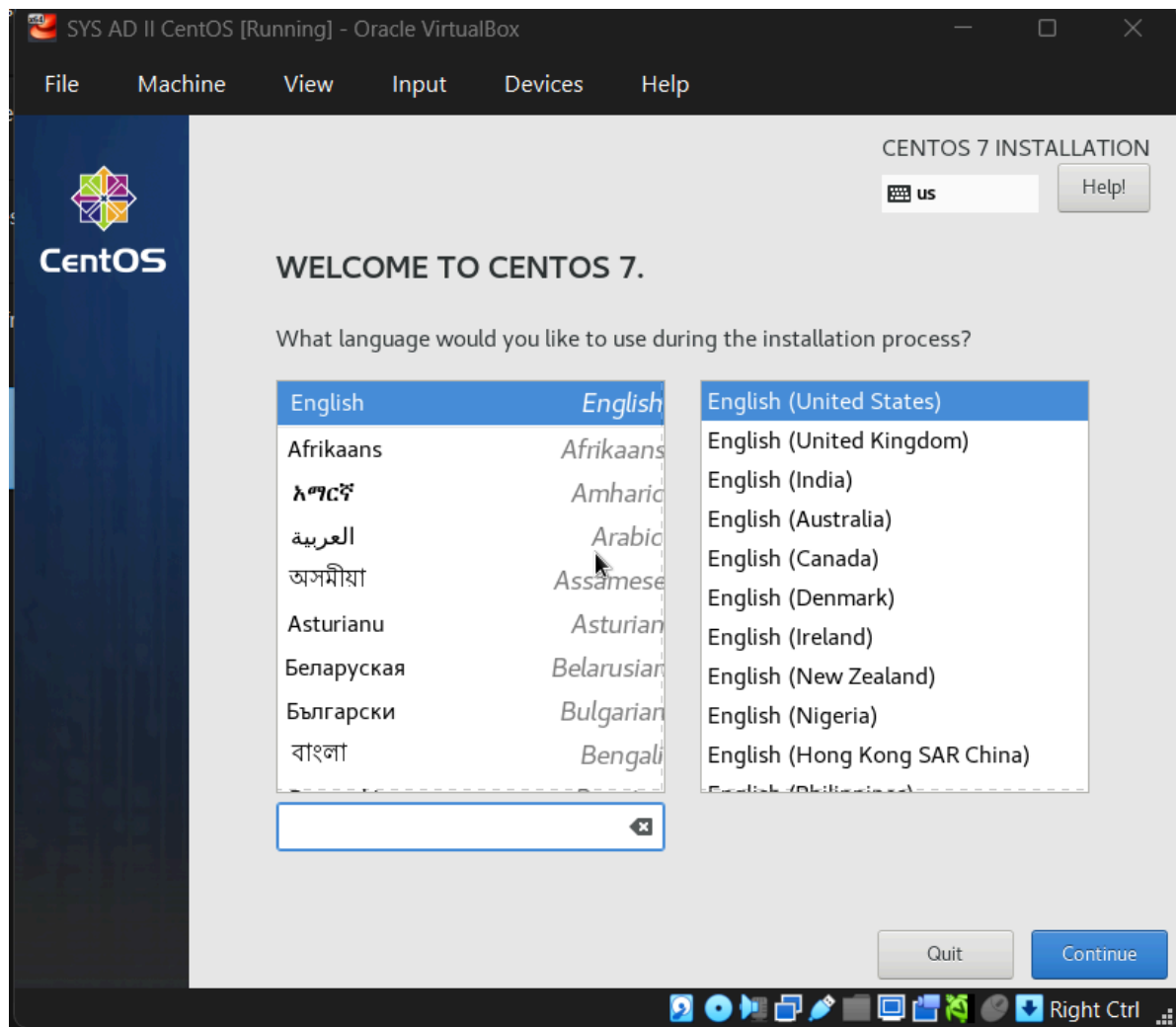
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Course/Section: CPE 212 - CPE31S1	Date Submitted: December 16, 2025
Instructor: Engr. Robin Valenzuela	Semester and SY: 2nd Sem - 2025-2026
Activity 3: Install SSH server on CentOS or RHEL 8	
1. Objectives: 1.1 Install Community Enterprise OS or Red Hat Linux OS 1.2 Configure remote SSH connection from remote computer to CentOS/RHEL-8	
2. Discussion: CentOS vs. Debian: Overview CentOS and Debian are Linux distributions that spawn from opposite ends of the candle. CentOS is a free downstream rebuild of the commercial Red Hat Enterprise Linux distribution where, in contrast, Debian is the free upstream distribution that is the base for other distributions, including the Ubuntu Linux distribution. As with many Linux distributions, CentOS and Debian are generally more alike than different; it isn't until we dig a little deeper that we find where they branch. CentOS vs. Debian: Architecture The available supported architectures can be the determining factor as to whether a distro is a viable option or not. Debian and CentOS are both very popular for x86_64/AMD64, but what other archs are supported by each? Both Debian and CentOS support AArch64/ARM64, armhf/armhfp, i386, ppc64el/ppc64le. (Note: armhf/armhfp and i386 are supported in CentOS 7 only.) CentOS 7 additionally supports POWER9 while Debian and CentOS 8 do not. CentOS 7 focuses on the x86_64/AMD64 architecture with the other archs released through the AltArch SIG (Alternate Architecture Special Interest Group) with CentOS 8 supporting x86_64/AMD64, AArch64 and ppc64le equally. Debian supports MIPSel, MIPS64el and s390x while CentOS does not. Much like CentOS 8, Debian does not favor one arch over another—all supported architectures are supported equally. CentOS vs. Debian: Package Management Most Linux distributions have some form of package manager nowadays, with some more complex and feature-rich than others. CentOS uses the RPM package format and YUM/DNF as the package manager. Debian uses the DEB package format and dpkg/APT as the package manager.	

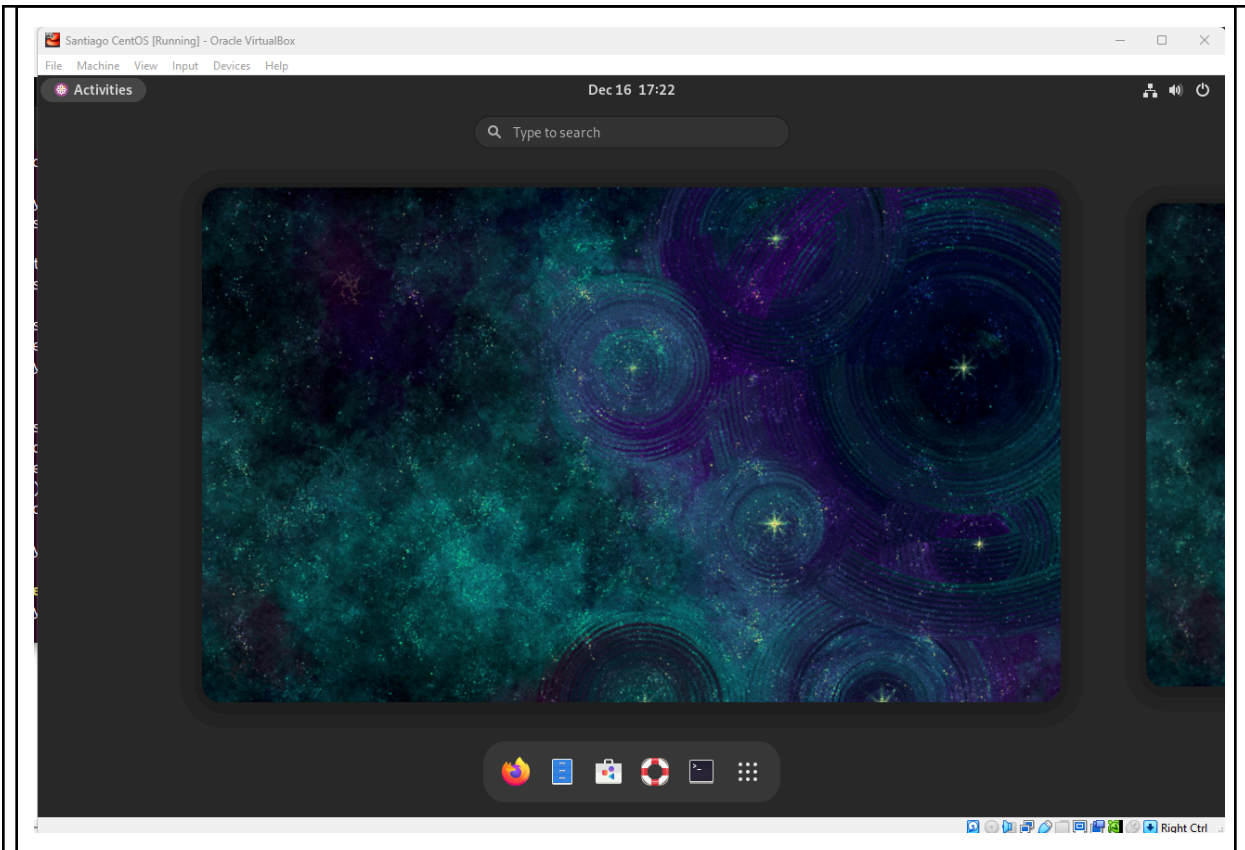
Both offer full-feature package management with network-based repository support, dependency checking and resolution, etc.. If you're familiar with one but not the other, you may have a little trouble switching over, but they're not overwhelmingly different. They both have similar features, just available through a different interface.

Task 1: Download the CentOS or RHEL-8 image (Create screenshots of the following)

1. Download the image of the CentOS here:
http://mirror.rise.ph/centos/7.9.2009/isos/x86_64/
2. Create a VM machine with 2 Gb RAM and 20 Gb HD.
3. Install the downloaded image.
4. Show evidence that the OS was installed already.

TASK 1





Task 2: Install the SSH server package *openssh*

1. Install the ssh server package *openssh* by using the *dnf* command:
\$ dnf install openssh-server
2. Start the *sshd* daemon and set to start after reboot:
\$ systemctl start sshd
\$ systemctl enable sshd
3. Confirm that the *sshd* daemon is up and running:
\$ systemctl status sshd
4. Open the SSH port 22 to allow incoming traffic:
\$ firewall-cmd --zone=public --permanent --add-service=ssh
\$ firewall-cmd --reload
5. Locate the ssh server man config file */etc/ssh/sshd_config* and perform custom configuration. Every time you make any change to the */etc/ssh/sshd-config* configuration file reload the *sshd* service to apply changes:
\$ systemctl reload sshd

TASK 2

```
jalene@vbox:~  
Running scriptlet: openssh-clients-9.9p1-3.el9.x86_64 2/6  
Running scriptlet: openssh-server-9.9p1-3.el9.x86_64 3/6  
Upgrading      : openssh-server-9.9p1-3.el9.x86_64 3/6  
Running scriptlet: openssh-server-9.9p1-3.el9.x86_64 3/6  
Running scriptlet: openssh-server-9.9p1-2.el9.x86_64 4/6  
Cleanup        : openssh-server-9.9p1-2.el9.x86_64 4/6  
Running scriptlet: openssh-server-9.9p1-2.el9.x86_64 4/6  
Running scriptlet: openssh-clients-9.9p1-2.el9.x86_64 5/6  
Cleanup        : openssh-clients-9.9p1-2.el9.x86_64 5/6  
Cleanup        : openssh-9.9p1-2.el9.x86_64 6/6  
Running scriptlet: openssh-9.9p1-2.el9.x86_64 6/6  
Verifying      : openssh-9.9p1-3.el9.x86_64 1/6  
Verifying      : openssh-9.9p1-2.el9.x86_64 2/6  
Verifying      : openssh-clients-9.9p1-3.el9.x86_64 3/6  
Verifying      : openssh-clients-9.9p1-2.el9.x86_64 4/6  
Verifying      : openssh-server-9.9p1-3.el9.x86_64 5/6  
Verifying      : openssh-server-9.9p1-2.el9.x86_64 6/6  
  
Upgraded:  
  openssh-9.9p1-3.el9.x86_64      openssh-clients-9.9p1-3.el9.x86_64  
  openssh-server-9.9p1-3.el9.x86_64  
  
Complete!  
[jalene@vbox ~]$
```

```
jalene@vbox:~ — systemctl status sshd  
Complete!  
[jalene@vbox ~]$ systemctl start sshd  
[jalene@vbox ~]$ systemctl enable sshd  
  
[jalene@vbox ~]$  
[jalene@vbox ~]$ systemctl status sshd  
● sshd.service - OpenSSH server daemon  
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: ena>  
   Active: active (running) since Tue 2025-12-16 18:13:24 PST; 1min 2s ago  
     Docs: man:sshd(8)  
           man:sshd_config(5)  
   Main PID: 3116 (sshd)  
     Tasks: 1 (limit: 22969)  
    Memory: 1.5M (peak: 1.8M)  
       CPU: 11ms  
    CGroup: /system.slice/sshd.service  
            └─3116 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"  
  
Dec 16 18:13:24 vbox systemd[1]: Starting OpenSSH server daemon...  
Dec 16 18:13:24 vbox sshd[3116]: Server listening on 0.0.0.0 port 22.  
Dec 16 18:13:24 vbox sshd[3116]: Server listening on :: port 22.  
Dec 16 18:13:24 vbox systemd[1]: Started OpenSSH server daemon.  
lines 1-16/16 (END)
```

```
jalene@vbox:~  
[jalene@vbox ~]$ systemctl status sshd  
● sshd.service - OpenSSH server daemon  
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: ena  
   Active: active (running) since Tue 2025-12-16 18:13:24 PST; 1min 2s ago  
     Docs: man:sshd(8)  
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   Main PID: 3116 (sshd)  
     Tasks: 1 (limit: 22969)  
    Memory: 1.5M (peak: 1.8M)  
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Dec 16 18:13:24 vbox systemd[1]: Starting OpenSSH server daemon...  
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Dec 16 18:13:24 vbox sshd[3116]: Server listening on :: port 22.  
Dec 16 18:13:24 vbox systemd[1]: Started OpenSSH server daemon.  
  
[jalene@vbox ~]$ firewall-cmd --zone=public --permanent --add-service=ssh  
Warning: ALREADY_ENABLED: ssh  
success  
[jalene@vbox ~]$ firewall-cmd --reload  
success  
[jalene@vbox ~]$
```

Task 3: Copy the Public Key to CentOS

1. Make sure that **ssh** is installed on the local machine.
2. Using the command **ssh-copy-id**, connect your local machine to CentOS.
3. On CentOS, verify that you have the **authorized_keys**.

TASK 3

```
jalene@workstation:~$ ssh-copy-id jalene@10.0.2.15  
The authenticity of host '10.0.2.15 (10.0.2.15)' can't be established.  
ED25519 key fingerprint is SHA256:4pJ+byL8wQZv4Mg+03Le/j40R0AcxuaXrBgl/KSsCw4.  
This host key is known by the following other names/addresses:  
  ~/.ssh/known_hosts:1: [hashed name]  
  ~/.ssh/known_hosts:4: [hashed name]  
  ~/.ssh/known_hosts:5: [hashed name]  
  ~/.ssh/known_hosts:6: [hashed name]  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out a  
ny that are already installed  
/usr/bin/ssh-copy-id: INFO: 2 key(s) remain to be installed -- if you are prompted no  
w it is to install the new keys  
jalene@10.0.2.15's password:  
  
Number of key(s) added: 2  
  
Now try logging into the machine, with: "ssh 'jalene@10.0.2.15'"  
and check to make sure that only the key(s) you wanted were added.
```

This is my first attempt at performing ssh on centos. However, it keeps on logging into the same ubuntu local machine. I resolved it by changing the network setting into Host-Only Adapter to change the CentOS ip address.

```
jalene@centos:~  
[jalene@centos ~]$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:38:91:df brd ff:ff:ff:ff:ff:ff  
    inet 192.168.56.104/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s3  
        valid_lft 575sec preferred_lft 575sec  
    inet6 fe80::a00:27ff:fe38:91df/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
[jalene@centos ~]$
```

Verification

```
jalene@workstation:~$ ssh jalene@192.168.56.104  
jalene@192.168.56.104's password:  
Activate the web console with: systemctl enable --now cockpit.socket  
  
Last login: Fri Jan  2 13:03:30 2026 from 192.168.56.102  
[jalene@centos ~]$ nano ~/.ssh/authorized_keys  
[jalene@centos ~]$ nano ~/.ssh/authorized_keys  
[jalene@centos ~]$ cat ~/.ssh/authorized_keys  
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIM5MkIPj1JeEB3TLz4TrhTUaR0YDrCLLNcdPNUcnQfd jalene@workstation  
[jalene@centos ~]$  
  
[jalene@centos ~]$ cat ~/.ssh/authorized_keys  
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIM5MkIPj1JeEB3TLz4TrhTUaR0YDrCLLNcdPNUcnQfd  
jalene@workstation  
[jalene@centos ~]$
```

Task 4: Verify ssh remote connection

1. Using your local machine, connect to CentOS using ssh.
2. Show evidence that you are connected.

TASK 4

```
jalene@centos:~  
jalene@workstation:~$ sudo nano /etc/hosts  
[sudo] password for jalene:  
jalene@workstation:~$ ssh jalene@192.168.56.104  
The authenticity of host '192.168.56.104 (192.168.56.104)' can't be established.  
ED25519 key fingerprint is SHA256:1nrtpXGkbf30G9wpe8KR3fqKky/1yuaDPd4E2BLIrz8.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.168.56.104' (ED25519) to the list of known hosts  
.  
jalene@192.168.56.104's password:  
Activate the web console with: systemctl enable --now cockpit.socket  
  
Last login: Fri Jan  2 12:47:48 2026  
[jalene@centos ~]$
```

Logging in without password

```
jalene@workstation:~$ ssh jalene@192.168.56.104  
Activate the web console with: systemctl enable --now cockpit.socket  
  
Last login: Fri Jan  2 13:10:31 2026 from 192.168.56.102  
[jalene@centos ~]$
```

Reflections:

Answer the following:

1. What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux distributions?

Choosing the best distribution between Debian and Red Hat still depends on the user's needs. In choosing a distribution, it is important that we consider our purpose first, our requirements, and our needs in a distribution. Debian-based systems are ideal for learning and general use, on the other hand, Red Hat-based systems are better for enterprise environments because they provide stronger and more advanced features which makes them reliable for production servers and corporate systems.

2. What are the main differences between Debian and Red Hat Linux distributions?

The main differences between the mentioned Linux distributions are their package managements, support models, and their target use. Debian uses APT with .deb packages and relies mainly on community support. Red Hat uses DNF/YUM with .rpm packages and focuses on enterprise environments.